

AI-Based Microscopy Image Analysis for Predicting Urinary Tract and Renal Diseases

Project Proposal Report

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ABSTRACT

Urine microscopy is a vital diagnostic tool for identifying renal and urinary tract diseases by analysing sediment particles such as RBCs, WBCs, casts, and crystals. However, its manual nature introduces subjectivity, variability, and inefficiencies, particularly in low-resource settings. This research proposes a comprehensive, AI-powered diagnostic system that integrates urine microscopy image analysis with patient metadata (e.g., age, symptoms, lifestyle) to enhance diagnostic accuracy and personalization. The system is designed to detect and classify key sediment components, including isomorphic and dysmorphic RBCs, WBCs, various types of crystals and casts, as well as bacteria categorized by morphology (cocci, bacilli, spirochetes). These findings are used to assist in the diagnosis of urinary tract infections (UTIs), hematuria, glomerulonephritis, and kidney stone formation, enabling a more comprehensive and clinically relevant assessment. By bridging image data with clinical context, this approach addresses key limitations of traditional urinalysis and advances the role of AI in diagnostic medicine.

Keywords: *Urine Microscopy, AI in Diagnostics, Urinary Tract Infections (UTIs), Hematuria, Glomerulonephritis, Kidney Stones, Red Blood Cells (RBCs), White Blood Cells (WBCs), Urinary Casts, Urinary Crystals, Bacteria Classification (Cocci, Bacilli, Spirochetes), Deep Learning, Patient Metadata Integration, Image-Based Diagnostics, Explainable AI, Medical Image Analysis, Multimodal AI System, Renal Disease Detection, Clinical Decision Support, Digital Urinalysis.*

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1. INTRODUCTION

1.1 Background Literature

Urine microscopy plays a pivotal role in the diagnosis and monitoring of various renal and urinary tract diseases. As one of the most common non-invasive diagnostic methods, microscopic examination of urine allows clinicians to identify and quantify critical components such as red blood cells (RBCs), white blood cells (WBCs), epithelial cells, casts, and crystals. These elements serve as biomarkers for a wide range of conditions, including infections, kidney stones, glomerulonephritis, and malignancies [1] [2]

Despite its clinical importance, urine microscopy is highly dependent on the expertise of laboratory technicians. The manual nature of the process makes it time-consuming, subjective, and prone to inter-observer variability [3] [4]. These limitations are particularly concerning in low-resource settings where experienced professionals and automated analyzers may be unavailable. As a result, there has been a significant push towards integrating advanced technologies such as Artificial Intelligence (AI) and Machine Learning (ML) to reduce human error, improve diagnostic consistency, and enhance scalability [5] [6].

AI has made notable advancements in several medical imaging disciplines, including radiology, dermatology, and pathology [7] [8]. However, its adoption in urinalysis, especially for urine sediment analysis, remains limited. Most existing studies focus on detecting a single type of sediment particle such as bacteria for UTI diagnosis [1] [3] [9] RBCs for hematuria [10], or crystals for kidney stones using convolutional neural networks (CNNs) or object detection models like YOLOv5 [11] [12]. While these models achieve respectable detection accuracies ranging from 88% to 94%, they lack the ability to simultaneously analyze multiple sediment types or to make meaningful connections between sediment composition and disease outcomes.

Furthermore, recent studies employing bidirectional context propagation networks [1], instance segmentation models [8] [13], and hybrid deep learning frameworks [14] [15] have demonstrated promising improvements in detection and classification performance. Nonetheless, they rarely incorporate patient metadata such as age, sex, past medical history, lifestyle habits, or reported symptoms [16] [17] [18] [19], which can significantly influence diagnostic interpretation. A multimodal AI system that combines image-based particle recognition with contextual patient information is urgently needed to achieve more accurate, holistic, and personalized diagnoses.

To address these gaps, our research proposes the development of a comprehensive, AI-powered diagnostic system that performs end-to-end urine sediment analysis. The system aims to detect and classify key particle types, WBCs, RBCs, casts, and crystals from microscopic images and combine these findings with patient metadata (e.g., age, symptoms, lifestyle) to improve disease prediction accuracy [6] [16] [19]. This integrated approach is expected to significantly enhance the diagnostic process and reduce reliance on manual interpretation.

White Blood Cells (WBCs) are immune cells that play a critical role in the body's defense against infections. In urinalysis, elevated WBC counts are indicative of inflammation or infection, commonly seen in conditions such as urinary tract infections (UTIs), interstitial

nephritis, and pyelonephritis [9] [16] [17]. Morphologically, WBCs appear as round cells with granular interiors and are larger than RBCs. The presence of WBC casts, in particular, is a strong indicator of upper UTIs such as pyelonephritis [2] [4]. Detecting WBCs accurately helps clinicians distinguish between bacterial infections and non-infectious inflammatory conditions.

Lower UTIs are generally confined to the bladder and urethra and present with symptoms such as dysuria, urinary frequency, and suprapubic pain. These infections are commonly caused by bacteria such as *Escherichia coli* (*E. coli*), which are rod-shaped and can be visualized under a microscope [9] [20]. In contrast, upper UTIs affect the kidneys and are more serious, often presenting with systemic symptoms like fever, chills, flank pain, and vomiting. WBC casts in urine microscopy are a hallmark of pyelonephritis and serve as a critical diagnostic marker [2] [18].

Red Blood Cells (RBCs) in urine signify hematuria, a condition that ranges from benign to serious depending on the source of bleeding [21] [15]. The morphology of RBCs provides diagnostic insight: isomorphic RBCs are uniform in shape and suggest bleeding from the lower urinary tract (e.g., bladder or urethra), whereas dysmorphic RBCs have irregular shapes and typically originate from the kidneys, indicating glomerular damage [5] [22]. Hematuria may result from infections, trauma, vigorous exercise, or malignancies like bladder cancer [19].

Dysmorphic RBCs are particularly important in diagnosing glomerulonephritis, an inflammatory condition of the glomeruli that often results in proteinuria and elevated serum creatinine. In contrast, isomorphic RBCs may point to benign causes such as exercise-induced hematuria or transient bleeding from the bladder wall due to catheterization. Thus, the ability to distinguish between RBC types is essential for making accurate differential diagnoses and avoiding unnecessary invasive procedures [15] [10].

Casts are cylindrical structures formed in the distal convoluted tubules and collecting ducts of the nephron. They consist of a protein matrix, primarily Tamm-Horsfall protein and may contain cellular elements [2] [4] [13]. Different cast types provide clues about specific renal conditions. RBC casts are strongly associated with glomerulonephritis, WBC casts indicate pyelonephritis or interstitial nephritis, granular casts suggest acute tubular necrosis (ATN), and waxy casts are indicative of chronic kidney disease or end-stage renal failure [14] [23].

Granular casts, especially the "muddy brown" variant, are one of the most characteristic findings in acute tubular necrosis, a condition that can result from ischemia, toxins, or sepsis. Waxy casts are seen in advanced renal failure and are formed from degenerated granular casts. They are broad, homogenous, and highly refractive, with a smooth texture. The accurate classification of these casts is critical not only for diagnosing renal diseases but also for staging disease severity and guiding treatment decisions [13] [23].

Crystals are solid aggregates of solutes that precipitate from urine. Their presence is often a precursor to kidney stone formation. Calcium oxalate crystals are the most prevalent and are associated with high oxalate intake, low fluid consumption, or metabolic disorders. Uric acid crystals form in acidic urine and may signal gout or excessive purine metabolism. Struvite crystals, typically seen in women, are linked to urinary tract infections caused by urease-

producing bacteria. Cystine crystals are rare and result from genetic disorders such as cystinuria [11] [22] [24].

Each crystal type has distinct morphological features under microscopy. Calcium oxalate crystals appear as envelope-shaped or dumbbell-like structures. Uric acid crystals are typically rhomboid or rosette-shaped. Struvite crystals, often referred to as "coffin-lid" crystals, are indicative of infection stones. Cystine crystals appear as hexagonal plates and are pathognomonic for cystinuria. Detecting and identifying these crystals early can lead to timely interventions and prevent the recurrence of painful and obstructive kidney stones [21].

Bacteria in urine samples are visualized as free-floating or clustered microorganisms and are a hallmark of urinary tract infections. They may present in various morphologies: cocci (spherical), bacilli (rod-shaped), and spirochetes (spiral-shaped) [9] [20]. *Escherichia coli* (*E. coli*), a rod-shaped bacterium, accounts for nearly 90% of UTIs. The presence of bacteria alongside WBCs and clinical symptoms such as dysuria, urgency, and fever confirms the diagnosis of infection. Additionally, identifying the shape and arrangement of bacteria (e.g., chains or clusters) can aid in targeting antibiotic therapy and improving patient outcome [17] [20].

To implement this diagnostic capability, we propose a multimodal AI model that combines convolutional image feature extraction with structured patient metadata. This approach will allow the model to contextualize visual findings with clinical information, thereby increasing diagnostic precision and relevance [16], [19]. If ethical approval is granted, the system will be trained on real clinical datasets. In its absence, synthetic metadata generated through domain knowledge will be used to simulate realistic patient profiles and train the model [15], [18].

In practical terms, the AI model will be deployed via a user-friendly web-based interface. This platform will enable clinicians and laboratory technicians to upload microscopy images and receive real-time diagnostic interpretations supported by AI. The system will also feature explainable AI modules that highlight detected features and provide reasoning for predictions, thereby building trust among healthcare providers and supporting informed decision-making [6], [15].

The proposed system also allows for scalability and modularity. As new particle types or disease markers are discovered, they can be integrated into the model framework without disrupting its existing functionality. This ensures long-term utility and adaptability in evolving medical environments [8] [25].

In conclusion, this research aims to revolutionize urine sediment analysis by introducing an AI-driven diagnostic platform that integrates image processing with patient metadata. By doing so, we address the key limitations of manual analysis, provide accurate and timely diagnosis, and contribute to the broader field of AI applications in clinical diagnostics.

1.2 Research Problem

Urine microscopy is a fundamental diagnostic tool for detecting urinary tract infections (UTIs), kidney stones, glomerulonephritis, and other renal diseases [1], [3], [9]. However, the current manual analysis process is labour-intensive, time-consuming, and highly dependent on the expertise of technicians, leading to significant variability and potential for human errors [2], [4]. These challenges are particularly pronounced in healthcare settings with limited resources, affecting diagnostic accuracy and timely patient care [6], [12].

While artificial intelligence (AI) and machine learning (ML) methods have demonstrated remarkable success in medical imaging domains such as radiology and pathology [7], [8], their application in urine sediment analysis remains limited and fragmented. Most prior research has focused on detecting individual urine sediment components such as bacteria for UTI diagnosis, or isolated detection of casts and crystals using models like CNNs and YOLOv5 [3], [9], [11], [22]. These methods have reported high detection accuracies but fail to provide an integrated solution capable of simultaneously identifying multiple sediment types and linking these findings directly to disease outcomes [13], [14].

Moreover, existing approaches often overlook the integration of critical patient metadata, such as age, medical history, lifestyle, and urinary symptoms, which are essential for personalized and clinically meaningful disease prediction [15], [16], [19]. Without combining image-based data with clinical context, diagnostic models risk reduced accuracy and clinical relevance [14], [18].

Therefore, there is a pressing need for an integrated AI system that can concurrently detect and classify a comprehensive range of urine sediment particles including white blood cells (WBCs), red blood cells (RBCs), casts, and crystals and utilize patient metadata to enhance early and precise prediction of urinary tract and renal diseases [13], [16], [19]. Such a system would automate laborious tasks, reduce the diagnostic workload, minimize human errors, and improve patient outcomes by delivering reliable, context-aware diagnostic support [14], [25].

1.3 Research Gap

A significant challenge in current automated urine sediment analysis research is the lack of integrated, multi-disease diagnostic systems. While many studies report high accuracy in detecting individual particle types like RBCs, WBCs, crystals, and epithelial cells (ranging from 85% to 98%) using CNN models such as YOLOv5, EfficientNet, and ResNet variants [3], [5], [6], [11], [12], they often operate in isolation. These models focus on single-particle detection tasks, limiting their ability to support real clinical decision-making across multiple diseases.

Despite some progress in particle classification, automated cast detection remains underexplored. For example, Li et al. proposed a method using ResNet50 and Feature Pyramid Networks for cast recognition [4], but this was limited to basic detection without subtype classification. Similarly, studies focusing on multi-crystal classification (e.g., calcium oxalate, uric acid, and struvite) failed to connect the results to higher-level disease prediction systems [22], [24].

Another major limitation is the poor integration of clinical metadata in model design. While some research investigates RBC morphology differentiating dysmorphic from isomorphic

types [10] these studies do not incorporate essential metadata such as age, trauma history, or urinary symptoms, which are vital for accurate hematuria diagnosis. Likewise, UTI-related models often emphasize bacterial presence or WBC count [1], [9], [16], [17] but do not consider cast information that is critical to differentiate upper and lower UTIs [2], [4], [23].

Explainable AI (XAI) is also lacking in this domain. Most models function as black boxes, offering minimal insight into how decisions are made. Although feature importance has been explored [14], [15], these methods are still basic and insufficient for clinical trust and transparency.

Dataset limitations further impact the field. Most publicly available datasets are small and lack diversity. For instance, the UMID dataset includes just 3,700 annotated samples across three categories [18]. Clinical models need much larger and varied datasets to ensure reliable performance across populations and imaging conditions. Additionally, inconsistent standards in imaging protocols, magnifications, and annotations across studies make it difficult to create generalizable models or conduct cross-institutional validations [6], [12].

Bacterial classification also remains weak, especially for Gram-positive bacteria and atypical morphologies, with many automated systems showing low sensitivity in these areas [9], [20]. Real-time deployment in clinical settings is another unmet challenge. Most AI models are designed for lab use and require high-end computing infrastructure, limiting their use in point-of-care settings where simplicity, speed, and affordability are crucial [25].

More importantly, current systems do not combine multiple findings to generate holistic disease predictions. For instance, detecting both RBC casts and dysmorphic RBCs could strongly indicate glomerulonephritis [10], [23], but no model currently integrates such findings for end-to-end disease prediction.

Furthermore, no existing system offers comprehensive clinical decision support that merges microscopic findings with patient metadata. There is no platform that can simultaneously predict conditions such as kidney stones, glomerulonephritis, pyelonephritis, and bladder cancer from a unified dataset of images and metadata [13], [19], [25]. The integration of such systems into real clinical workflows is rarely studied, and the potential for reducing manual workload in laboratories remains largely unexamined.

To address these critical gaps, the proposed research introduces a novel AI-powered framework that combines deep learning image analysis with structured clinical metadata to enable comprehensive, real-time multi-disease prediction. This includes explainability features to build clinician trust, standardized imaging and annotation protocols, and a large-scale diverse dataset to improve generalizability. By shifting from isolated classification tasks to a unified diagnostic decision support system, this research aims to advance the field and create clinically viable, scalable solutions.

1.4 Research Significance

Area of Significance	Description	Key Reference
Automation of Urine Microscopy	Reduces manual effort, inter-observer variability, and diagnosis delays through AI-based image analysis.	[1] [3] [9] [14]
Multi-particle Detection	Simultaneous identification of WBCs, RBCs (isomorphic/dysmorphic), bacteria, casts, and crystals.	[1], [4], [5], [6], [7]
UTI Classification	Differentiates between upper UTI (WBC casts), lower UTI , and no UTI using sediment patterns	[3], [5], [14], [15]
Kidney Stone Prediction	Crystal type classification (e.g., calcium oxalate, struvite) predicts type of nephrolithiasis.	[10], [21], [24]
RBC Morphology for Hematuria Origin	Differentiates glomerular vs. non-glomerular bleeding (dysmorphic vs. isomorphic RBCs).	[6], [13], [16]
Cast Identification and Disease Mapping	RBC/WBC/granular/waxy casts linked to glomerulonephritis, ATN, CKD, pyelonephritis.	[4], [5], [12], [13]
Integration of Clinical Metadata	Enhances model accuracy using patient context (age, symptoms, diagnosis tags).	[13], [14], [17]
Real-time Diagnostic Tool	Web-based interface allows technicians to upload and receive instant AI-based diagnosis suggestions.	[13], [17], [18]
Support for Resource-Limited Settings	Ideal for hospitals with limited lab expertise or high patient volume.	[1], [11], [14]

2. OBJECTIVES

2.1 Main Objective

To develop an AI-powered, end-to-end diagnostic support system that classifies key urinary sediment particles from microscopy images, integrates patient metadata, and predicts associated urinary tract and renal diseases. The system will provide risk assessment and explainable clinical insights through a web-based platform, supporting early and accurate diagnosis in real-world healthcare environments.

2.2 Specific Objectives

- Design and implement deep learning models (e.g., CNNs, YOLOv5 variants) to predict, classify, and quantify key urinary particles, including red blood cells (RBCs), white blood cells (WBCs), urinary casts, crystals, and bacteria from high-resolution urine microscopy images.
- Perform morphological analysis of RBCs to distinguish between isomorphic and dysmorphic cells, enabling differentiation between glomerular and non-glomerular hematuria origins (e.g., trauma, bladder cancer, nephropathy).
- Predict and classify diagnostically relevant cast types (RBC, WBC, granular, waxy) and estimate cast density per high-power field to support diagnosis of glomerulonephritis, pyelonephritis, ATN, and CKD.
- Accurately identify and characterize urinary crystals (e.g., calcium oxalate, uric acid, struvite, cystine) using shape, texture, and clustering features to assess the likelihood of kidney stone formation.
- Classify bacterial forms based on shape and distribution (e.g., cocci, bacilli) and quantify WBCs to stratify UTIs into upper UTI (pyelonephritis), lower UTI, or non-infectious findings.
- Develop a metadata-enhanced prediction pipeline that combines image-based features with structured patient metadata (age, sex, symptoms, lifestyle habits, medical history) using late fusion or joint learning strategies.
- Train multimodal neural networks or hybrid ensemble classifiers to estimate disease probability scores and risk levels, optimizing for both sensitivity and specificity.
- Build a real-time, web-based clinical interface that supports image uploads, displays particle detection outputs, and delivers AI-driven diagnostic suggestions with confidence scores.
- Integrate patient-level dashboards for individualized reporting and population-level insights for research and clinical monitoring (e.g., frequency of specific sediment types across patient groups).

3. METHODOLOGY

3.1 System Architecture Diagram

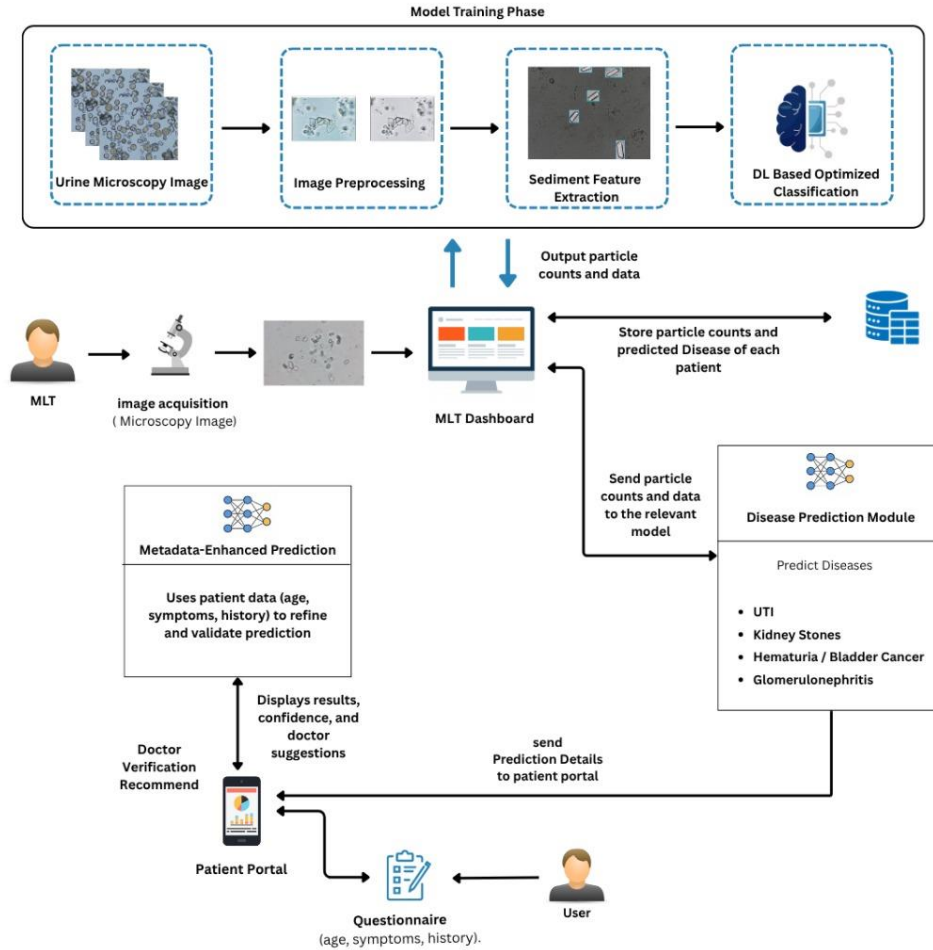


Figure 1: System Architecture Diagram

3.2 Description

To effectively implement the proposed AI-powered diagnostic system for urinary sediment analysis and disease prediction, a robust and modular methodology has been adopted, covering the development, training, and deployment of deep learning models capable of detecting and classifying key urinary particles such as RBCs, WBCs, casts, crystals, and bacteria from high-resolution microscopy images. The system also integrates structured patient metadata such as age, symptoms, lifestyle habits, and medical history through multimodal learning to enhance diagnostic accuracy and personalize disease predictions. The methodology follows a structured pipeline comprising data acquisition, image preprocessing, model development, metadata fusion, validation, and deployment via a real-time web interface. This comprehensive approach ensures clinical reliability, scalability, and usability in real-world laboratory environments.

while supporting the early detection of conditions such as UTIs, kidney stones, glomerulonephritis, and bladder cancer through explainable AI-driven insights.

The architecture of the proposed system is illustrated in **Figure 1** and comprises a modular and scalable pipeline that automates the process of urinary sediment analysis, metadata integration, and disease prediction. It begins with the acquisition of high-resolution urine microscopy images using laboratory-grade microscopes equipped with digital imaging sensors. Medical Laboratory Technicians (MLTs) capture images representing key urinary sediment particles such as red blood cells (RBCs), white blood cells (WBCs), urinary casts, crystals and bacteria. These images are then subjected to a preprocessing stage where techniques such as contrast enhancement, histogram equalization, denoising, and resizing are applied to normalize image quality across samples, minimize device-induced inconsistencies, and prepare the data for model input.

Once preprocessing is complete, sediment feature extraction is carried out using computer vision and deep learning techniques to isolate morphological attributes such as particle shape, size, texture, boundary definition. The extracted features are fed into a deep learning-based classification module composed of YOLOv5 and hybrid CNN architectures, which are trained to accurately detect and categorize particles into clinically meaningful groups. These include isomorphic and dysmorphic RBCs for hematuria analysis, four cast types (RBC, WBC, granular, and waxy) for kidney condition assessment, crystal subtypes (e.g., calcium oxalate, uric acid, struvite, cystine) for kidney stone risk estimation, and WBCs or bacterial forms (e.g., cocci, bacilli) for UTI prediction. Each particle type is identified and counted per high-power field (HPF) and stored for diagnostic inference.

Following classification, an initial diagnostic report is automatically generated and displayed on the MLT interface. This report includes preliminary predictions for conditions such as urinary tract infections, kidney stones, glomerulonephritis, and hematuria-related cancers, including the prediction of hematuria causes by differentiating between glomerular and non-glomerular bleeding based on RBC morphology. Importantly, no patient metadata is required at this stage. Instead, the system prompts the patient to access their report through a secure web portal and encourages them to submit personal health information such as age, gender, urinary symptoms, hydration status, and relevant medical history via a structured form. This patient-driven metadata submission model reduces the burden on MLTs while improving prediction specificity.

Once the metadata is received, it is combined with the image-based features using a multimodal fusion model. This model employs a late fusion strategy enhanced with attention mechanisms to generate more personalized and accurate predictions. For example, it enables the system to distinguish glomerular versus non-glomerular hematuria based not only on RBC morphology but also on risk factors such as trauma or smoking history. The refined report, including updated disease probabilities, confidence scores, and personalized suggestions, is then made available to the patient through their dashboard. If any critical indicators are detected, the system can recommend further doctor review or medical follow-up.

The entire workflow is supported by a full-stack web application that allows secure image uploads, real-time report generation, and two-way communication between patients and clinical staff. The platform includes advanced features for historical tracking, side-by-side image comparisons, and progressive disease monitoring over time, ensuring the system is clinically usable and adaptable for long-term laboratory deployment. This architecture not only streamlines the diagnostic process but also incorporates patient feedback into the AI loop, promoting greater transparency, precision, and scalability in real-world healthcare environments.

4. COMMERCIALIZATION

4.1 Target Audience and Market Space

The primary target audience includes:

- Hospitals and Clinical Laboratories seeking to automate and standardize urine microscopy diagnostics.
- Diagnostic Chains and Pathology Labs aiming to improve operational efficiency and reduce reliance on human interpretation.
- Rural and Low-Resource Healthcare Facilities where skilled technicians and advanced diagnostic equipment are lacking.
- Telemedicine and Mobile Health Providers that require scalable, cloud-ready diagnostic tools for remote care delivery.
- HealthTech Companies and Medical AI Vendors looking for integration-ready AI modules in nephrology and urology domains.

The market space aligns with the rapidly growing sectors of AI in healthcare, digital pathology, and clinical decision support systems (CDSS). These sectors are increasingly driving the adoption of intelligent diagnostic platforms that offer real-time, explainable, and cost-effective solutions for improving diagnostic accuracy and clinical efficiency.

5. EXPECTED CONTRIBUTIONS

- Development of a comprehensive AI-based diagnostic system capable of detecting and classifying key urine sediment particles such as WBCs, RBCs (isomorphic and dysmorphic), casts, crystals, and bacteria (cocci, bacilli, spirochetes) from microscopic images.
- Integration of patient metadata (age, symptoms, lifestyle, medical history) with image-based findings to enhance disease prediction accuracy for conditions such as UTIs, glomerulonephritis, and kidney stones.
- A multimodal model that offers holistic disease assessment rather than isolated particle identification.
- A web-based, user-friendly interface enabling clinicians and lab technicians to upload images and receive real-time, explainable AI-driven diagnostic feedback.
- Improved model generalizability through the use of a large, diverse dataset and standardized image preprocessing methods.
- Validation of the system's performance using key metrics such as accuracy, sensitivity, specificity, and F1-score.

6. RESEARCH SIGNIFICANCE

- This research addresses a critical need for automation in urine microscopy, especially in resource-limited settings where skilled personnel and advanced tools are scarce.
- By integrating AI with both image analysis and patient metadata, the system promotes personalized, context-aware diagnostics, reducing false positives/negatives and enhancing clinical decision-making.
- The introduction of explainable AI features fosters trust among clinicians by offering transparency in model decisions, a crucial factor for adoption in healthcare environments.
- Real-time diagnostic capability via a web platform has the potential to significantly reduce diagnostic turnaround times and workload for laboratory staff.
- The system lays the foundation for scalable, modular clinical decision support tools that can evolve to include additional biomarkers or disease targets, promoting long-term impact and relevance in digital healthcare.
- Ultimately, the solution contributes to more accurate, efficient, and equitable diagnosis of renal and urinary tract conditions, improving patient outcomes and healthcare delivery

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